

Campus Marketplace

Sprint 2 Planning Document

Software Design 2026

Prepared: April 2026

Sprint Duration: 2 Weeks

1. Sprint 2 Overview

Sprint 2 builds on the foundation established in Sprint 1. The focus of this sprint is to deliver a complete buyer experience, in-app messaging, user dashboards and profiles for all roles, and a seller rating system. Additionally, this sprint addresses gaps from Sprint 1 by properly implementing user dashboards and profiles for Student, Staff, and Admin roles.

Sprint Goals

- Deliver complete user dashboards and profiles for Student, Staff, and Admin roles
- Implement seller rating and review system (US7)
- Enable buyers to initiate purchase and trade offers (US8)
- Build real-time buyer-seller chat (US9)
- Implement trade facility slot booking (US10)
- Complete rating and review submission (US16)
- Resolve Sprint 1 gaps: properly wire up AdminUsers.jsx, AccessDenied.jsx, and CreateListing

Sprint 2 Story Summary

Story ID	Title	Priority	Status
US-D1	Student Dashboard & Profile	High	To Do
US-D2	Staff Dashboard & Profile	High	To Do
US-D3	Admin Dashboard & Profile	High	To Do
US-S1	Sprint 1 Gap Fix: Wire up CreateListing, AdminUsers, AccessDenied	High	To Do
US7	View Seller Rating	Medium	To Do
US8	Initiate Purchase or Trade Offer	High	To Do
US9	Buyer-Seller Chat	Medium	To Do
US10	Book Drop-Off Slot	Medium	To Do
US16	Leave Rating and Review	Medium	To Do

2. User Dashboards & Profiles

US-D1 — Student Dashboard & Profile

As a student, I want a personalised dashboard showing my listings, active transactions, and messages so that I can manage my marketplace activity in one place.

Tasks

Task	Assignee	Status	Notes
Build StudentDashboard.jsx page	TBD	To Do	Route: /dashboard
Display user profile info (name, photo, email, role)	TBD	To Do	Fetch from Firestore
Show student's active listings with edit/delete	TBD	To Do	Filter by user UID
Show active transactions (buying and selling)	TBD	To Do	
Show unread message count badge	TBD	To Do	
Build StudentProfile.jsx viewable by other users	TBD	To Do	Route: /profile/:uid
Display average rating and review count on profile	TBD	To Do	

User Acceptance Tests

Given	When	Then
A logged-in student opens their dashboard	The page loads	They see their profile info, active listings, and transaction summary
A buyer views a seller listing	They click the seller's name	The seller's public profile opens showing rating and listings

US-D2 — Staff Dashboard & Profile

As trade facility staff, I want a dashboard showing upcoming drop-offs, collections, and pending confirmations so that I can efficiently manage facility operations.

Tasks

Task	Assignee	Status	Notes
Build StaffDashboard.jsx page	TBD	To Do	Route: /staff/dashboard
Display upcoming drop-offs and collections	TBD	To Do	Fetch from Firestore transactions
Add Confirm Drop-Off and Confirm Collection buttons	TBD	To Do	
Show outstanding cash shortfall per transaction	TBD	To Do	
Build StaffProfile.jsx with role badge	TBD	To Do	
Restrict staff dashboard to staff/admin roles only	TBD	To Do	Use useRequireRole

User Acceptance Tests

Given	When	Then
A logged-in staff member opens their dashboard	The page loads	They see upcoming drop-offs and collections with confirm buttons
A student tries to access /staff/dashboard	They navigate to the URL	They are shown the AccessDenied page

US-D3 — Admin Dashboard & Profile

As an admin, I want a comprehensive dashboard showing platform statistics, user management, and moderation tools so that I can oversee the entire marketplace.

Tasks

Task	Assignee	Status	Notes
Build AdminDashboard.jsx as the admin home page	TBD	To Do	Route: /admin/dashboard
Show platform stats (total users, listings, transactions)	TBD	To Do	
Link to User Management (AdminUsers.jsx)	TBD	To Do	Already exists
Link to reported content moderation queue	TBD	To Do	
Build AdminProfile.jsx with admin badge	TBD	To Do	
Restrict admin dashboard to admin role only	TBD	To Do	Use useRequireRole

User Acceptance Tests

Given	When	Then
A logged-in admin opens their dashboard	The page loads	They see platform stats and links to user management and moderation
A student tries to access /admin/dashboard	They navigate to the URL	They are shown the AccessDenied page

3. US7 — View Seller Rating

As a student buyer, I want to view a seller's rating and transaction history so that I can assess their trustworthiness before buying.

Tasks

Task	Assignee	Status	Notes
Display average star rating on seller profile page	TBD	To Do	Aggregate from reviews collection
Show total review count on seller profile	TBD	To Do	
Show completed transaction count on seller profile	TBD	To Do	
Link seller name on listing card to their profile	TBD	To Do	Route: /profile/:uid
Build star rating display component (read-only)	TBD	To Do	Reuse in US16

User Acceptance Tests

Given	When	Then
A buyer views a listing	They click on the seller's name	The seller's profile opens showing their average rating and completed transaction count
A seller has no reviews yet	A buyer views their profile	The profile shows 'No reviews yet' and a rating of 0

4. US8 — Initiate Purchase or Trade Offer

As a student buyer, I want to initiate a purchase or trade offer on a listing so that I can begin the transaction process.

Tasks

Task	Assignee	Status	Notes
Add Buy Now button on listing detail page	TBD	To Do	Visible to non-owners only
Add Make Trade Offer button on listing detail page	TBD	To Do	Only for For Trade listings
Create transaction document in Firestore on offer initiation	TBD	To Do	Status: pending
Notify seller of new offer via in-app notification	TBD	To Do	Firestore notification document
Allow seller to accept or decline the offer	TBD	To Do	Update transaction status
Build listing detail page (ViewListingDetail.jsx)	TBD	To Do	Route: /listing/:id

User Acceptance Tests

Given	When	Then
A logged-in buyer views a For Sale listing	They click Buy Now	A transaction is created with status pending and the seller receives a notification
A seller receives a trade offer	They click Accept	The transaction status updates to Accepted and the buyer is notified
A seller receives an offer they don't want	They click Decline	The transaction status updates to Declined and the buyer is notified
The listing owner views their own listing	The page loads	The Buy Now and Make Trade Offer buttons are not visible

5. US9 — Buyer-Seller Chat

As a student, I want to chat with a buyer or seller through the platform so that I can negotiate and arrange transaction details safely without sharing personal contact information.

Tasks

Task	Assignee	Status	Notes
Design Chat.jsx UI with message list and input box	TBD	To Do	Route: /chat/:transactionId
Implement real-time chat using Firestore messages subcollection	TBD	To Do	onSnapshot listener
Restrict chat access to the two parties in the transaction	TBD	To Do	Check buyerId and sellerId
Add unread message badge on dashboard	TBD	To Do	Count unread messages
Mark messages as read when chat is opened	TBD	To Do	
Add link to chat from transaction detail page	TBD	To Do	

Firestore Data Structure

transactions/{transactionId}/messages/{messageId}

- senderId: string (Firebase UID)
- content: string
- timestamp: Firestore timestamp
- isRead: boolean

User Acceptance Tests

Given	When	Then
A buyer initiates an offer and a transaction is created	Either party opens the chat	Both parties can send and receive messages in real time
A student tries to access a chat they are not part of	They navigate to the chat URL	They are denied access and shown an error
A user receives a new message while on their dashboard	The page is open	An unread message badge updates in real time

6. US10 — Book Drop-Off Slot

As a student, I want to book a drop-off time slot at the trade facility so that I can securely hand over an item I have sold.

Tasks

Task	Assignee	Status	Notes
Build SlotBooking.jsx page showing available time slots	TBD	To Do	Route: /book-slot/:transactionId
Fetch available slots from Firestore facility config	TBD	To Do	
Allow only students in an accepted transaction to book	TBD	To Do	Check transaction status
Store booking in Firestore and prevent double-booking	TBD	To Do	Use Firestore transactions
Send booking confirmation notification to student	TBD	To Do	Firestore notification
Display booked slot on transaction detail page	TBD	To Do	

User Acceptance Tests

Given	When	Then
A student has an accepted transaction	They select an available slot and confirm	The slot is booked and a confirmation notification is sent
A slot is already fully booked	A student tries to book it	The slot shows as unavailable and cannot be selected
A student without an accepted transaction tries to book	They access the booking page	They are redirected with an error message

7. US16 — Leave Rating and Review

As a student, I want to leave a star rating and written review after a completed transaction so that others can make informed decisions about trading with that person.

Tasks

Task	Assignee	Status	Notes
Prompt both parties to rate after transaction is marked complete	TBD	To Do	Show review prompt on dashboard
Build ReviewForm.jsx with star rating selector and text input	TBD	To Do	1-5 stars
Store review in Firestore reviews collection	TBD	To Do	Linked to transaction and reviewer
Recalculate and update average rating for reviewed user	TBD	To Do	Update user document
Display reviews list on user profile page	TBD	To Do	Most recent first
Prevent duplicate reviews per transaction	TBD	To Do	Check if review exists

User Acceptance Tests

Given	When	Then
A transaction is marked complete	Both parties are prompted to leave a review	Each can submit a star rating and written review
A user receives a new review	The review is submitted	Their average rating is updated immediately on their profile
A user tries to review the same transaction twice	They attempt to submit a second review	The form shows an error and the duplicate is not saved

8. Technical Notes & Architecture

Frontend Routes for Sprint 2

Route	Component	Access
/dashboard	StudentDashboard.jsx	Authenticated students
/staff/dashboard	StaffDashboard.jsx	Staff & Admin only
/admin/dashboard	AdminDashboard.jsx	Admin only
/profile/:uid	StudentProfile.jsx	Public
/listing/:id	ViewListingDetail.jsx	Public
/chat/:transactionId	Chat.jsx	Transaction parties only
/book-slot/:transactionId	SlotBooking.jsx	Accepted transaction parties
/review/:transactionId	ReviewForm.jsx	Completed transaction parties